MLFF ADAPT-MON1 fixed common online monitor specification

Status: implemented in mdstats 0.20.123a0

1. Scope

ADAPT-MON1 replaces variable fold-local checkpoint-monitor inputs in newly prepared MLFF campaigns with two immutable, campaign-common online validation artifacts. It changes monitor construction and lineage only. It does **not** implement adaptive stopping, lightweight finalist ranking, or retirement of EVAL-MF; those remain later gates.

The online monitors are model-selection evidence only and must never supply gradients.

2. Canonical policy

`OnlineMonitorPolicy` uses schema `mdstats.online-monitor-policy.v1` and defaults to:

- target budget: **256 configurations**;
- replay budget: **512 configurations**;
- deterministic monitor seed: **161803**;
- target strategy: `balanced_condition_run_time_systematic`;
- replay strategy: `chemistry_size_systematic`.

Changing any policy field changes the policy digest and therefore the materialization/protocol identity.

3. Common target monitor

The target parent domain is DATA5 `outer_monitor` evidence only. Fold-local training, fold-local validation, and locked/held-out test roles are not eligible parents.

The selector:

1. groups eligible units by declared condition and run identity;
2. distributes the requested budget as evenly as possible across those strata;
3. orders frames by source trajectory/time within each stratum;
4. uses a deterministic random-start systematic sample to spread selections across time; and
5. records exact ordered configuration identities, source-frame indices, stratum counts, requested/realized size, seed, strategy, parent digest, and true-label domain.

Every fold, seed, training method, and final-development job prepared under the same label domain receives the **same target monitor membership and**

artifact content digest. This is required so online errors are directly comparable across competing runs.

If fewer than 256 eligible configurations exist, all eligible configurations are used and the fallback is explicitly recorded.

4. Independent true-label replay monitor

The replay parent must be a `ReplayFileArtifact` with `ReplayLabelMode.TRUE_DFT`. Foundation pseudo labels may remain replay **training** data, but they cannot define the absolute online replay validation metric.

Replay selection uses a stable ordering over exact composition/chemistry, atom count/size bin, and source order followed by deterministic systematic sampling. The selected 512-frame default subset is materialized at:

```
shared/replay/online_true_replay_monitor.xyz
```

The materialized artifact is immediately re-inspected. Geometry identity, ordered membership, label identity, and true-label mode must match the parent selection exactly or preparation fails. If fewer than 512 true-label configurations exist, all available configurations are used and the fallback is recorded.

For multi-head replay jobs:

- MACE `pt_train_file` remains the configured replay training artifact; and
- MACE `pt_valid_file` is the fixed true-label online replay monitor.

Thus validation evidence is independent of the gradient-label mode.

5. Identity contracts

`OnlineMonitorRecord` uses schema `mdstats.online-monitor-record.v1`. A record binds:

- role (`target` or `replay`);
- parent-domain digest;
- policy digest;
- requested and realized sizes;
- exact ordered selected identities;
- source indices;
- per-stratum available/selected counts;
- strategy and seed;
- label mode and parent role; and
- explicit fallback reason codes.

New DATA8 preparation uses `mdstats.data8-preparation-bundle.v2`. New production materialization uses `mdstats.production-materialization-plan.v4`. New training protocol identities use `mdstats.training-protocol-identity.v4` when online-monitor evidence is present. The protocol binds the policy digest, target-record digest, replay-record digest, and materialized replay-valid artifact digest as one quartet.

Historical DATA8/production-plan/protocol schemas remain readable. Low-level callers that do not request ADAPT-MON1 retain legacy construction semantics for compatibility, while new campaign CLI preparation always binds the new monitor policy and requires independent true replay labels.

6. Leakage and role invariants

- Online monitor configurations never contribute gradients.
- Target monitor selection is restricted to the common DATA5 outer-monitor domain.
- Locked/sealed test evidence cannot be promoted into the online monitor role.
- Target-training geometries may not overlap the true-label replay online monitor by exact geometry identity.
- Monitor membership is selected once and reused; it is not redrawn per epoch, fold, or seed.
- The monitor seed is identity, not an informal reproducibility hint.

7. Precision boundary

ADAPT-PREC1 remains authoritative. `single` jobs evaluate these monitors with the FP32 learned model; `double` jobs use the FP64 learned model. mdstats-owned SSE/RMSE/statistical accumulation remains hard-coded FP64 under either model precision.

8. Current runtime boundary

ADAPT-MON1 itself freezes the common validation artifacts. Beginning in 0.20.124a0, ADAPT-STOP1 consumes their already-paid force-RMSE rows at every evaluated epoch and can terminate training at the target-success, replay-exhaustion, or hard-epoch boundary without additional monitor inference. ADAPT-EVAL1 has not yet retired the existing mixed-fidelity evaluation stage.

9. Acceptance tests

The gate requires regression coverage for:

1. deterministic target and replay membership regeneration;
2. exact 256/512 defaults and explicit small-parent fallbacks;
3. common target membership across every fold/final job;
4. target condition/run/time coverage rather than first-N selection;
5. TRUE_DFT replay enforcement and exact label/geometry round-trip;

6. no target-training/replay-monitor geometry overlap;
7. MACE `pt_valid_file` separation from replay training data;
8. protocol/digest serialization and corruption rejection; and
9. historical schema readability.